

Task 2: General questions

1a-

$$\text{Input} = Z(x_1, x_2) = W_1x_1 + W_2x_2 + b \quad W_1 = 2 \quad W_2 = -1 \quad b = 0.5$$

$$Z(2, 4) = 2(2) + (-1)(4) + 0.5 = 0.5 > 0 \quad \text{class label} = 1$$

$$Z(4, 2) = 2(4) + (-1)(2) + 0.5 = 6.5 > 0 \quad \text{class label} = 1$$

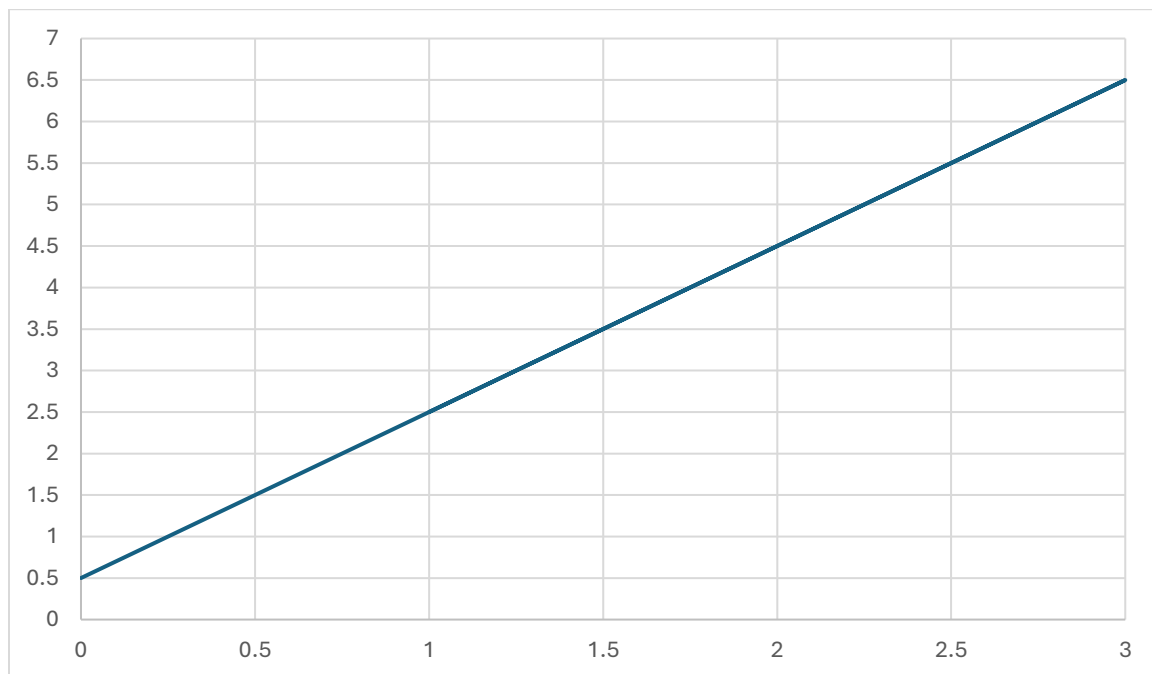
$$Z(2, 2) = 2(2) + (-1)(2) + 0.5 = 2.5 > 0 \quad \text{class label} = 1$$

b-

Boundary at $Z = 0$

$$2x_1 - x_2 + 0.5 = 0$$

$$x_2 = 2x_1 + 0.5$$



C-

Limitations of Perceptron Algorithm:

Only works for linearly separable data

- It can only solve problems where the two classes can be separated by a straight line (or hyperplane).
- It cannot solve problems like XOR, where the classes are mixed in a nonlinear way.

Might never stop training if data is not separable

- If the data cannot be perfectly separated, the algorithm may **keep updating forever** without reaching a final solution.

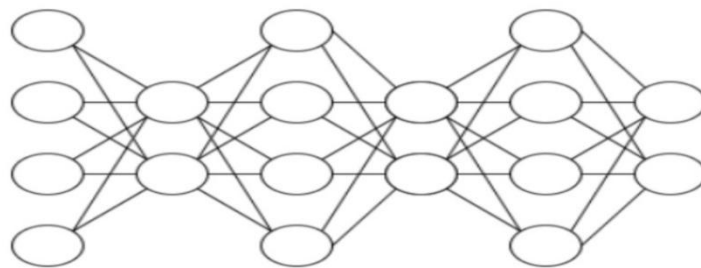
No confidence level

- It only gives a hard answer (0 or 1).
- It doesn't tell how confident it is about the prediction.

Can't handle complex patterns

- Since it has no hidden layers, it cannot learn complicated or curved decision boundaries.

2-



Number of neurons = 18

Number of hidden layers = 4

Learnable parameters:

$$(4+1) * 2 + (2+1) * 4 + (4+1) * 2 + (2+1) * 4 + (4+1) * 2 = 54$$

Total Learnable parameters = 54

3-Consider the following 1-d array [3, 4, 6, 3, 2, 3, 4] apply convolution with the following 1-d filter [1, 2, -1] and 0 padding.

$$[3,4,6]: 3*1 + 4*2 + 6*-1 = 5$$

$$[4,6,3]: 4*1 + 6*2 + 3*-1 = 13$$

$$[6,3,2]: 6*1 + 3*2 + 2*-1 = 10$$

$$[3,2,3]: 3*1 + 2*2 + 3*-1 = 4$$

$$[2,3,4]: 2*1 + 3*2 + 4*-1 = 4$$

Output : [5,13,10,4,4]

4-For a 1-d array with a length of L and a filter of size F , what is the length of the output array? How much padding would be required to maintain a constant output size? Explain your answer.

Length of Output array: $L-F+1$

Padding required: since the size of the output with padding will be: $L + 2p - k + 1$

So the amount of padding required would be $2p - k + 1 = 0$

$$P = (k-1)/2$$

5- For the shown 2d array apply the following 2x2 convolution filter, with a stride of 1:

Array:

[[3, 5, 2, 2],
[1, -1, 3, -1],
[0, 6, 3, 7],
[10, -3, 5, 6]]

Filter:

[[2, 1],
[-1, 2]]

$$\left\lfloor \frac{n+2p-f}{s} + 1 \right\rfloor \times \left\lfloor \underbrace{\frac{n+2p-f}{s}} + 1 \right\rfloor$$

Output feature map = [3,3]

Top left 2x2: $(3*2)+(5*1)+(1*-1)+(-1*2)=6+5-1-2=8$

Top middle 2x2: $(5*2)+(2*1)+(-1*-1)+(3*2)=10+2+1+6=19$

Top right 2x2: $(2*2)+(2*1)+(3*-1)+(-1*2)=4+2-3-2=1$

Middle left 2x2: $(1*2)+(-1*1)+(0*-1)+(6*2)=2-1+0+12=13$

Center 2x2: $(-1*2)+(3*1)+(6*-1)+(3*2)=-2+3-6+6=1$

Middle right 2x2: $(3*2)+(-1*1)+(3*-1)+(7*2)=6-1-3+14=16$

Bottom left 2x2: $(0*2)+(6*1)+(10*-1)+(-3*2)=0+6-10-6=-10$

Bottom middle 2x2: $(6*2)+(3*1)+(-3*-1)+(5*2)=12+3+3+10=28$

Bottom right 2x2: $(3*2)+(7*1)+(5*-1)+(6*2)=6+7-5+12=20$

Output Map function:

8	19	1
13	1	16
10	28	20